astradand

ASTRADAND gesture wand: manufacturing manual

> **Status: design only; nothing built or tested on hardware.** PCB: **order-ready: every automatic check passes**. Printed shell: generated, never printed. Firmware: engine host-tested, board code compiled only. Expect to correct this manual after the first 5-unit prototype run.

1. What is made

A 232 mm hilt: two FDM-printed shell halves holding a 146 x 26 mm PCB (ATmega328P, MPU-6050 IMU, six WS2812B pixels, piezo, haptic motor), three AAA cells in line, a diffuser crystal, trigger cap, tip and pommel; a laser-cut display stand, a carton, nine spell cards. See `astradand/docs/CONCEPT.md` for play and `DESIGN` notes there.

2. Deliverables

Item	Where	State
PCB (Gerbers, BOM, CPL, KiCad, STEP)	`utility10/pcb/release/astradand/`	**order-ready: every automatic check passes**
Firmware	`astradand/firmware/release/astradand_8MHz.hex`; source `astradand/firmware/lib/ad_core.c`, `src/main_astradand.cpp`	engine 83 host checks; board code compiled (19.7 KB flash, 1.1 KB RAM)
Shell STL	`astradand/cad/out/hilt_upper.stl`, `hilt_lower.stl`, `crystal.stl`, `trigger_cap.stl`, `tip_cap.stl`, `pommel.stl`	watertight; unprototyped
Stand (laser)	`astradand/cad/out/stand_laser.svg/.dxf`	6 mm MDF; unprototyped
Carton dieline	`astradand/cad/out/carton_dieline.svg/.dxf`	inside 250 x 64 x 54 mm
Spell cards	`astradand/spellbook/out/spell_cards_A4.pdf` (+ one SVG per card)	Hindi names unreviewed
Assembly section	`astradand/cad/out/assembly_section.png`	drawing
Test plan / economics	`astradand/docs/TEST_PLAN.md`, `economics.md`	

3. Circuit board bill of materials

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
U1	ATmega328P-AU	TQFP-32_7x7mm_P0.8mm	ATMEGA328P-AU	C14877	SMT (JLC)
C1	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C2	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C3	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
R7	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
J2	ISP	PinHeader_2x03_P2.54mm_Vertical	ZHOURI 2.54-2*3	C5116479	hand/THT
J1	BATT 3xAAA	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
SW1	Power	SW_SPDT_PCM12	C&K PCM12SMTR	C221841	SMT (JLC)
Q2	AO3401A	SOT-23	AO3401A	C15127	SMT (JLC)
C4	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
D11	WS2812B-Mini	LED_WS2812B-Mini_PLCC4_3.5x3.5mm	Worldsemi WS2812B-Mini 3535 (LCSC listing; confirm the variant suffix against the datasheet)	C114583	SMT (JLC)
C20	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
D12	WS2812B-Mini	LED_WS2812B-Mini_PLCC4_3.5x3.5mm	Worldsemi WS2812B-Mini 3535 (LCSC listing; confirm the variant suffix against the datasheet)	C114583	SMT (JLC)
C21	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
D13	WS2812B-Mini	LED_WS2812B-Mini_PLCC4_3.5x3.5mm	Worldsemi WS2812B-Mini 3535 (LCSC listing; confirm the variant suffix against the datasheet)	C114583	SMT (JLC)
C22	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
D14	WS2812B-Mini	LED_WS2812B-Mini_PLCC4_3.5x3.5mm	Worldsemi WS2812B-Mini 3535 (LCSC listing; confirm the variant suffix against the datasheet)	C114583	SMT (JLC)
C23	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
D15	WS2812B-Mini	LED_WS2812B-Mini_PLCC4_3.5x3.5mm	Worldsemi WS2812B-Mini 3535 (LCSC listing; confirm the variant suffix against the datasheet)	C114583	SMT (JLC)
C24	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
D16	WS2812B-Mini	LED_WS2812B-Mini_PLCC4_3.5x3.5mm	Worldsemi WS2812B-Mini 3535 (LCSC listing; confirm the variant suffix against the datasheet)	C114583	SMT (JLC)
C25	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
U2	MPU-6050	InvenSense_QFN-24_4x4mm_P0.5mm	MPU-6050	C24112	SMT (JLC)
C10	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C11	2.2n	C_0603_1608Metric	Samsung CL10B222KB8NNNC X7R	C16033	SMT (JLC)
C12	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
C13	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
R20	4.7k	R_0603_1608Metric	0603 4.7k 1%	C23162	SMT (JLC)
R21	4.7k	R_0603_1608Metric	0603 4.7k 1%	C23162	SMT (JLC)
U3	AP2112K-3.3	SOT-23-5	Diodes AP2112K-3.3TRG1 (from the LCSC datasheet link)	C51118	SMT (JLC)
C5	1u	C_0603_1608Metric	0603 1uF X7R	C15849	SMT (JLC)
C6	1u	C_0603_1608Metric	0603 1uF X7R	C15849	SMT (JLC)
D20	SS14	D_SMA	Starsea SS14 SMA	C84343	SMT (JLC)
Q3	AO3401A	SOT-23	AO3401A	C15127	SMT (JLC)
Q4	MMBT3904	SOT-23	MMBT3904	C20526	SMT (JLC)
R33	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
R34	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
C7	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
R30	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
R31	330k	R_0603_1608Metric	UNI-ROYAL 0603WAF3303T5E	C23137	SMT (JLC)
R32	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
BZ1	PKLCS1212E4001	Buzzer_Murata_PKLCS1212E	PKLCS1212E4001-R1	C113159	SMT (JLC)
SW2	Cast trigger	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)
Q1	MMBT3904	SOT-23	MMBT3904	C20526	SMT (JLC)
D2	1N4148WS	D_SOD-323	1N4148WS	C2128	SMT (JLC)
R6	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
J4	Haptic motor	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT

Bought-in and mechanical (estimates for 2,000 units)

3 x AAA in-line holder with leads (JST-PH 2-pin), coin vibration motor with JST-PH 2-pin lead (3 V, 8-10 mm, under 80 mA), M2.5 x 8 mm self-tapping screws x 12, rubber grommet for the battery lead, translucent natural PETG or clear resin for the crystal, PETG (opaque, any dark colour) for the shell, 350 gsm board for the carton, 6 mm MDF for the stand, 300 gsm card for the spell cards.

4. Make the board

1. Order bare PCBs and SMT assembly per `manuals/STANDARD_PROCEDURES.md` A and B. The WS2812B-Mini, MPU-6050 (QFN, check the 0.5 mm pitch exposed-pad variant in the assembly preview), AP2112K, piezo and passives are on the JLC BOM when released. **The MPU-6050 is a not-recommended-for-new-designs part: confirm stock before ordering.**
2. Solder the hand/THT parts: ISP header J2, JST battery J1 and motor J4, slide switch SW1 (SMT), trigger SW2.
3. Check for shorts between +4V5, +3V3, VLED and GND before power.

5. Program

```
avrdude -p m328p -c usbasp -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFD:m
avrdude -p m328p -c usbasp -U flash:w:astradand/firmware/release/astradand_8MHz.hex:i
```

Pin map: D2 IMU INT, D3 trigger, D4 haptic, D5 LED supply switch, D6 WS2812 data, D9/D10 piezo, A0 battery divider, A4/A5 I2C. Then run Gate 1 of `astradand/docs/TEST_PLAN.md` before assembling any unit: it checks I2C, the WS2812 timing on a scope, sleep current and battery reading.

6. Print the shell (FDM prototype)

Part	File	Size (mm)	Notes
hilt_upper	`hilt_upper.stl`	232.0 x 35.6 x 17.8	volume 29.84 cm3
hilt_lower	`hilt_lower.stl`	232.0 x 35.6 x 17.8	volume 40.27 cm3
crystal	`crystal.stl`	51.4 x 6.8 x 12.4	volume 4.29 cm3
trigger_cap	`trigger_cap.stl`	8.4 x 8.4 x 7.7	volume 0.24 cm3
tip_cap	`tip_cap.stl`	22.0 x 34.0 x 34.0	volume 6.65 cm3
pommel	`pommel.stl`	19.8 x 33.0 x 33.0	volume 11.19 cm3

- Material: PETG for the shell (dark colour so the LEDs do not show through the wall), **translucent natural PETG or clear resin for the crystal**; nozzle 0.4 mm, layers 0.2 mm, 3 perimeters, 25 % infill, no supports for the halves (print each half with the split face on the bed, i.e. the open side down) except under the trigger hole bridge.
- The crystal needs a smooth top face: print with the long face down at 0.12 mm layers, sand 400-1000 grit and flame-polish lightly or finish with clear resin.
- Print one set first; test-fit the real PCB, battery tube and motor: the rails, bosses and the 0.3 mm tube clearance are first-pass numbers.

7. Assemble a unit

1. Seat the PCB (components up) on the rails of the upper half; the six pixels must sit under the crystal slit; the trigger button under the 6 mm hole.
2. Insert the three cells into the holder, route the leads to J1; seat the holder in the lower half's tube; press the motor into its pocket near J4 and plug it in (a drop of hot glue).
3. Place the diffuser crystal in the slit, add the trigger cap; close the halves and fasten the 12 screws (M2.5 x 8 into the pilot holes, 0.3 Nm): **the battery door is the shell itself, screw-closed**.
4. Press on the tip cap and pommel (friction fit, a dot of cyanoacrylate if loose).
5. Switch on: power-on tune, mana crystals lit.

8. End-of-line test (every unit)

Programme and verify; I2C scan; gyro rest bias under 5 dps; draw a circle and a Z (both must cast); colour sweep on all six pixels; sound and haptic; battery reading 4.5 V +-0.2; sleep current under 100 uA after 45 s; record on `manuals/templates/eol_test_sheet.md`.

9. Stand, carton, cards

- Stand: laser-cut `stand_laser.dxf` from 6 mm MDF (kerf test first), glue the two cradles into the base slots; seal with water-based lacquer; felt feet.
- Carton: print artwork on 350 gsm board, die-cut from `carton_dieline.dxf`, fold on the blue lines, glue the 14 mm flap.
- Cards: print `spell_cards_A4.pdf` on 300 gsm card at 100 %, cut on the crop marks (63 x 88 mm), round the corners. **Proofread the Devanagari names with a native speaker before printing.**

10. Packaging, compliance, limits

Label 9+ (small parts, battery), include a battery warning and a screw-closed compartment statement; Legal Metrology marks; verify toy-standard applicability, battery and EMC testing, LED eye safety (diffused, unmeasured) and e-waste obligations with a lab before selling (Gate 4). Not done: any hardware test, human gesture data, certification. Economics: marginal at Rs 2,499, target Rs 2,999 (`economics.md`).